

Dual-Reconfigurable Polarization and Pattern Wearable Antenna

Guo-Ping Gao, Jun-Jie Xia, Xuan Wang, Wen-Di Guo, Yan-Fei Zhang and Bin Hu

Abstract— With the development of wearable wireless systems and wireless body area networks, the 3.5 GHz WiMAX band has become critical for body-centric communications, creating an urgent demand for high-performance wearable antennas with multifunctional integration, compact size and stable on-body performance. This paper proposes a novel polarization and pattern reconfigurable wearable antenna for the 3.5 GHz WiMAX band. The antenna consists of two square FR-4 dielectric substrates and three metal layers. It can operate in omnidirectional mode, horizontal polarization mode and $\pm 45^\circ$ linear polarization modes. It has a compact size of $0.36 \times 0.36 \times 0.035 \lambda_0^3$, with a measured peak gain of up to 7.22 dBi. The antenna maintains stable performance near the human body, and its low specific absorption rate values verify its safety in practical wearable applications.

Index Terms—Patch antenna, pattern reconfigurable, polarization reconfigurable, wearable antenna.

I. INTRODUCTION

As a core component of wearable wireless systems and Wireless Body Area Network (WBAN), the performance of wearable antennas directly determines the communication quality of devices [1-3]. Fig. 1 depicts the application scenario of a human body equipped with a reconfigurable wearable antenna engaging in WBAN communications. Reconfigurable antennas are a new type of antenna that can dynamically adjust parameters such as frequency, radiation pattern and polarization through external control [4][5]. This enables a single antenna to meet communication requirements across multiple scenarios. Pattern reconfigurable antennas tune radiation beam properties using RF switches and related components, thus improving communication stability between wearable devices and base stations [6-9]. Polarization reconfigurable antennas achieve polarization switching via feeding phase tuning, radiating structure modification and tunable elements, thus effectively enhancing the anti-interference performance of communication links. Pattern and polarization reconfigurable antennas feature high functional integration [10-13]. They are usually compact or flexible and integrable with novel flexible

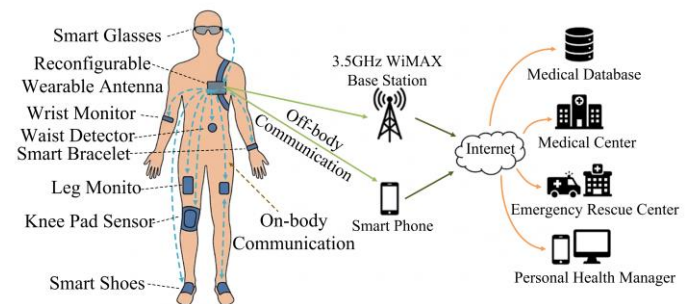


Fig. 1. Application scenario of WBAN communications.

and transparent materials, providing key technical support for high-performance wearable wireless systems [14-18].

In recent years, polarization and radiation pattern dual-reconfigurable antennas have garnered growing attention [19]. Ref. [20] presents a liquid-based reconfigurable dielectric resonator antenna with dual-mode and dual-polarization reconfigurability, but suffers from cumbersome control and low gain. Ref. [21] proposes a metasurface antenna with reconfigurable polarization and modes, yet its multilayer structure results in complex switches and degraded gain. Ref. [22] develops a dual-polarization mode reconfigurable antenna, whose application is restricted by complicated feeding networks. Ref. [23] introduces a multi-mode vertically polarized dielectric resonator antenna, but its high profile hinders integration. Therefore, the design of low-profile, multi-mode and high-gain reconfigurable antennas is of great research significance.

This paper proposes a novel a polarization and pattern reconfigurable wearable antenna for the 3.5 GHz WiMAX band for wearable body area network applications. Based on a differential feeding architecture. The feeding network excites omnidirectional and directional modes respectively, and shorting vias are used to tune the two modes to the same frequency point, realizing pattern reconfigurability. Benefiting from the differential feeding design, the antenna maintains a compact size of $0.36 \times 0.36 \times 0.035 \lambda_0^3$, a measured peak gain up to 7.22 dBi, and stable on-body radiation performance. Finally, simulated and measured results are in good agreement, and its low SAR values confirm excellent radiation performance and reliable safety for wearable applications.

II. ANTENNA DESIGN

A. Antenna Configurations

Fig. 2 shows the structural diagram of a wearable circular patch antenna with reconfigurable polarization and radiation pattern. The antenna structure comprises two square FR-4 dielectric substrates with side length W , a relative permittivity of 4.4 and a loss tangent of 0.02, as well as three metal layers

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Digital Object Identifier

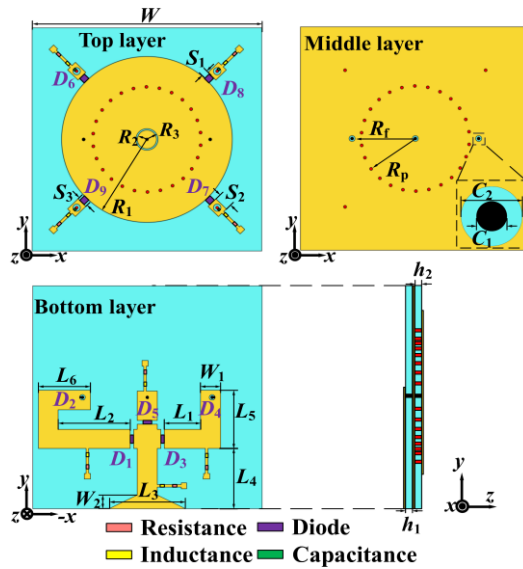


Fig. 2. Geometry of the proposed antenna.

including the top radiating patch, the ground plane and the bottom feeding line. The top radiating patch consists of a central circular patch and parasitic metal sheets distributed along its $\pm 45^\circ$ directions. The central feeding point of the circular patch excites the antenna's omnidirectional mode, and the left and right feeding points excite its directional mode. K equidistant shorting vias are introduced at a distance of R_p from the center of the circular patch to adjust the resonant frequencies of the two modes. The central circular patch is connected to the grounded parasitic metal sheets via diodes along its $\pm 45^\circ$ directions respectively. Circular holes are etched at the feeding metal via positions on the metal ground plane, which ensures the feeding process is free from the influence of the metal ground plane. The feeding layer adopts microstrip feeding. The final optimized structure parameters are given as follows: $R_1 = 23.2$, $R_2 = 2.6$, $R_3 = 3$, $R_f = 17.2$, $R_p = 14.7$, $C_1 = 0.9$, $C_2 = 2$, $h_1 = 1.5$, $h_2 = 1.5$, $S_1 = 1.8$, $S_2 = 4$, $S_3 = 3$, $L_1 = 10.2$, $L_2 = 20.2$, $L_3 = 17.6$, $L_4 = 17.4$, $L_5 = 15.8$, $L_6 = 14.6$, $W = 31.2$, $W_1 = 5.6$, $W_2 = 2$ (Units: mm).

B. Step to Realize the Proposed Antenna

Fig. 3 depicts the antenna's design process. We integrate the directional radiation structure Ant. A with the omnidirectional radiation structure Ant. B, and then add the reconfigurable feeding layer shown in Fig. 2 to the antenna's bottom, thus constructing the initial pattern-reconfigurable antenna Ant. 1. Feeding at points 2+ and 2- excites the TM_{11} mode of the circular patch antenna to achieve directional radiation, while feeding at point 1 excites the TM_{01} mode to realize omnidirectional radiation. A slot is etched at the patch center to prevent current from flowing back to the feeding layer under the directional mode. However, the two modes have different orders and resonant frequencies, so Ant. 1 cannot

TABLE I

OPERATING STATES OF THE ANTENNA

Operating state	Operating mode	States of diodes ($D_1 \sim D_9$)	Radiation Pattern	Polarization Direction
State 1	TM_{01}	000010000	Non-directional	/
State 2	TM_{11}	111101100	Directional	$\varphi = -45^\circ$
State 3	TM_{11}	111100011	Directional	$\varphi = +45^\circ$

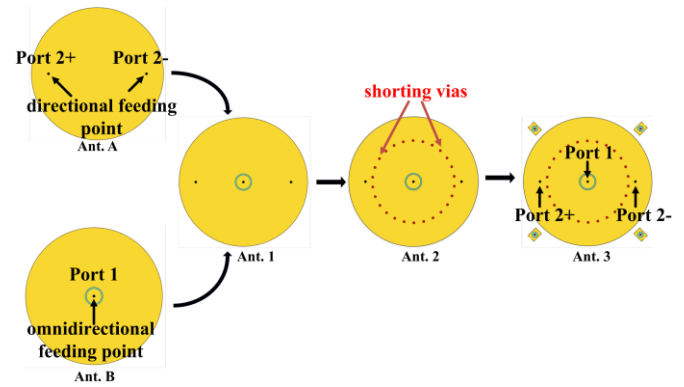


Fig. 3. Step to realize the proposed antenna.

meet the reconfigurable design requirements. Accordingly, Ant. 2 is equipped with k equidistant shorting vias, which are placed at a distance of R_p from the circular patch center to adjust the resonant frequencies of the two modes. The shorting vias effectively shorten the current path of the omnidirectional mode and increase its resonant frequency, finally enabling both modes to operate in the same frequency band. Ant. 3 adds grounded parasitic patches in different directions, and diodes control the connection between these parasitic patches and the central circular metal patch. This introduces asymmetric interference, which changes the antenna's polarization direction and breaks the limitation of the horizontal polarization mode, thus achieving polarization reconfigurability ultimately.

Table I summarizes the operating states of this polarization and pattern reconfigurable wearable antenna, and Fig. 4 shows the top and bottom current distributions of the antenna under different states. As presented in Table I, the antenna has three operating states in total.

When D_5 conducts with all other diodes off, the antenna operates in State 1, the RF signal transmits directly from the middle feeding line of the feeding layer to the top radiating layer, exciting the circular patch's TM_{01} mode to realize omnidirectional radiation. When $D_1 \sim D_4$ conduct and D_5 is off, the RF signal is fed to the top patch from the directional feeding lines on both sides of the feeding layer; the asymmetric structure causes current phase delay on the two feeding lines, producing a differential feeding-like effect that excites the top circular patch's TM_{11} mode to realize directional radiation (State 4). In this mode, the antenna works in State 2 with D_6 and D_7 conducting, achieving -45° linear polarization, and in State 3 with D_8 and D_9 conducting, achieving $+45^\circ$ linear polarization.

C. Parameter Selection

As mentioned in the design concept of this structure, the shorting vias arranged circularly and equidistantly on the circular patch are the key to tuning the two modes to the same frequency band. To achieve this goal, the key is to select a reasonable shorting via position R_p . Fig. 5(a) and (b) shows the influence of changing parameter R_p on the antenna's S_{11} parameter. As shown in Fig. 5(a), the current path of the antenna in omnidirectional mode elongates with the increase of R_p , leading to a decrease in the resonant frequency of the TM_{01} mode; as shown in Fig. 5(b), the resonant current path in directional mode shortens with the increase of R_p , resulting in an increase in the resonant frequency of the TM_{11} mode.

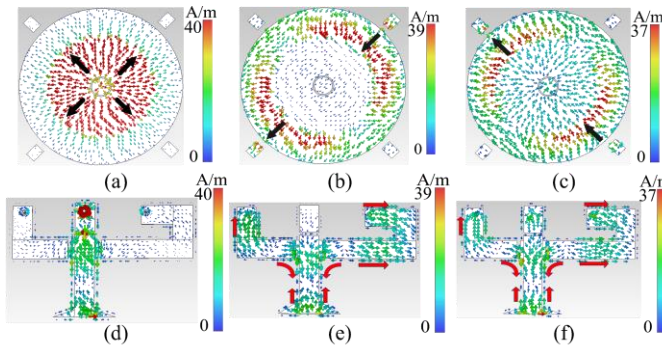


Fig. 4. Current distribution of the patch and feed line. (a) and (d) State 1. (b) and (e) State 2. (c) and (f) State 3.

Shorting vias are arranged exactly between the omnidirectional and directional feeding metal vias, thus R_p exerts entirely different effects on the patch surface current paths under the two modes. For the identical circular patch, the resonant frequency of the TM_{11} mode is consistently higher than that of the TM_{01} mode. The incorporation of shorting vias elevates the resonant frequency of the TM_{01} mode while lowering that of the TM_{11} mode, and $R_p=14.7$ mm is ultimately selected to tune the two modes to the same resonant frequency point.

The parameter selection of the triangular impedance matching modules at the bottom of the antenna's feeding layer is also crucial to the antenna's S_{11} parameter. Changing the length and height of the triangles alters the impedance of the microstrip transmission line. Selecting appropriate length and height of the triangles ensures good impedance matching within the same frequency band for the antenna in all operating states. Fig. 5(c) and (d), 5(e) and (f) show the influences of parameters L_3 and W_2 on the antenna's S_{11} parameter, respectively. As shown in Fig. 5(c) and (d), the impedance matching effect under both modes improves as L_3 decreases. When $L_3 = 17.6$ mm, the resonant frequency point of the omnidirectional mode drops below -10 dB, and that of the directional mode reaches the lowest value. As shown in Fig. 5(e) and (f), the impedance matching under both modes is also significantly improved as W_2 decreases. Therefore, $W_2 = 2$ mm is selected as the final structural parameter.

III. SIMULATED AND MEASURED RESULTS

Fig. 6 gives the simulated and measured S_{11} parameters of the proposed antenna. Results of all antenna operating states show increasing frequency. All operating bands fall within the 3.5 GHz WiMAX target band. The antenna realizes good impedance matching in this band and exhibits excellent overall performance. The measured S_{11} remains stable near the human body. Fig. 6(d) shows the fabricated antenna sample.

Fig. 7 shows current distributions at extra resonant point near the operating band for State 2 and 3. Compared with Fig. 4 (b) and (c), the extra resonance, which originates from mode splitting of the outer degenerate mode induced by perturbation of the differential feeding structure, deviates from the main band and hardly affects the main-mode excitation and reconfigurability.

Fig. 8 shows the simulated and measured normalized radiation patterns of the antenna. It can be observed from Fig.

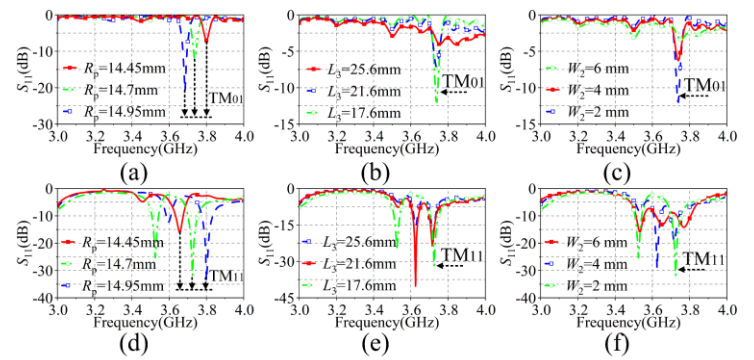


Fig. 5. The influence of key parameters on the antenna's S_{11} . (a) and (d) R_p . (b) and (e) L_3 . (c) and (f) W_2 .

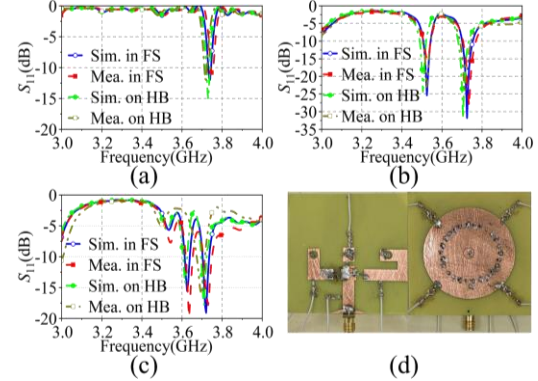


Fig. 6. Simulated and measured S_{11} parameters. (a) State 1. (b) State 2. (c) State 3. (d) Antenna sample.

8 (a) and (d) that the antenna presents favorable omnidirectional radiation performance in State 1. As shown in Fig. 8(b), (e), (c) and (f), the main beam of the antenna is nearly oriented along the $+z$ direction in State 2 and State 3, with excellent directional radiation characteristics. Overall, the measured results agree well with the simulated ones. The slight discrepancies may arise from the limitations of the measurement environment and antenna fabrication errors.

Fig. 9 presents the simulated and measured maximum gains of the proposed antenna in the 3.65-3.85 GHz frequency band under three operating states. Measurements are implemented in free space (FS) and on porcine tissue (PT), where PT is adopted to mimic human skin tissue. For State 1, the simulated maximum gain is 6.96 dBi, the measured gain in FS is 7.22 dBi, and the measured gain on PT is 6.44 dBi. For State 2, the simulated maximum gain is 7.97 dBi, the measured gain in FS is 6.86 dBi, and the measured gain on PT is 6.58 dBi. For State 3, the simulated maximum gain is 7.62 dBi, the measured gain in FS is 6.93 dBi, and the measured gain on PT is 6.28 dBi. Thus, the proposed design can be considered to offer good wearability.

To simulate the performance of the antenna on the human body surface, a human tissue simulation model is designed in this paper. The model comprises three layers: skin, fat and muscle.

TABLE II
HUMAN TISSUE PARAMETERS

	ϵ_r	σ (S/m)	Density(kg/m ³)	Thickness(mm)
Skin	37.6	1.52	1200	1.7
Fat	5.4	0.12	900	12
Muscle	52.3	1.84	1000	18

TABLE III
COMPARISON OF THE PROPOSED ANTENNA AND OTHER ANTENNAS

Reference	Modes	Polarizations	Switches	Overall Size/ λ_0^3	Peak Gain (dBi)	Band Width (GHz)
[20]	3	2	2	$0.62^2 \times 0.18$	4.1/4.9/5.0	2.4-2.8
[21]	6	3	16	$0.65^2 \times 0.067$	3.66/3.58/4.38 4.29/3.02/3.05	4.6-5.4
[22]	6	6	8	$1.32^2 \times 0.07$	6.3	3.68-3.86
[23]	9	8	9	$\pi \times 0.16^2 \times 0.14$	4/5.6/5.4	2.62-2.82
[24]	3	3	4	$1.35^2 \times 0.04$	5.5/3.0	5.12-5.20
[25]	5	5	2	$\pi \times 0.75^2 \times 0.06$	3.1/5.0/4.0 5.5/5.1	3.2-3.6
This work	3	2	9	$0.36^2 \times 0.035$	7.22/6.86/6.93	3.68-3.74

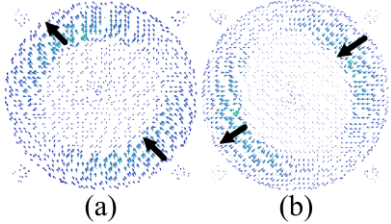


Fig. 7. Current distribution at lower resonant frequency. (a) State 2. (b) State 3.

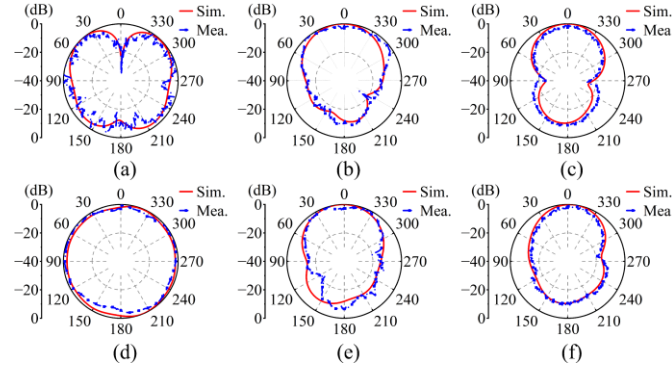


Fig. 8. Measured and simulated radiation patterns at 3.75GHz. (a) State 1, xoz -plane. (b) State 2, xoz -plane. (c) State 3, xoz -plane. (d) State 1, yo -plane. (e) State 2, yo -plane. (f) State 3, yo -plane.

The electromagnetic parameters of each layer are given in Table II. In the on-body performance simulation, as shown in Fig. 9(g), the antenna is placed 10 mm above the model to match real application scenarios. Fig. 10 shows the simulated specific absorption rate (SAR) results of the antenna, with the input power uniformly set to 0.1 W. In State 1, the maximum SAR values for 1g and 10g tissues are 0.1964 W/kg and 0.0769 W/kg, respectively. In State 2, the maximum SAR values for 1g and 10g tissues are 0.2259 W/kg and 0.1062 W/kg, respectively. In State 3, the maximum SAR values for 1g and 10g tissues are 0.2546 W/kg and 0.0978 W/kg, respectively. All values are much lower than the maximum SAR limits specified in the C95.1-2019 and EN 62209-3 standards, which proves the safety of the antenna for wearable applications.

Table III lists the performance of the proposed antenna and previously reported reconfigurable antennas in terms of reconfigurability, complexity, polarization modes, size, and gain. Some designs, such as that in Ref. [25], adopt four groups of mechanical rotation structures for mode control. Although this reduces the number of diodes, it results in higher system complexity in practical applications. The antenna proposed in this work features relatively low design complexity. While the number of diodes is not significantly reduced, the proposed design maintains a compact structure and achieves better radiation performance via differential

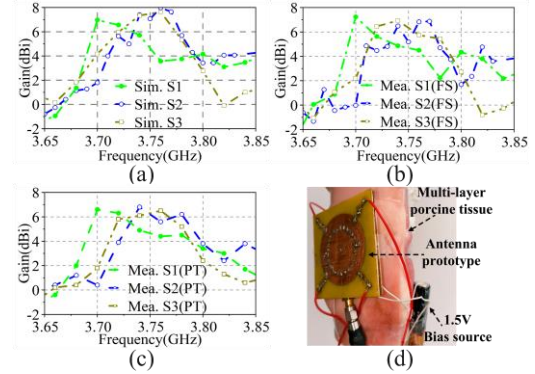


Fig. 9. (a) Simulated in free space. (b) Measured in free-space. (c) Measured on porcine tissue. (d) Measurement setup on porcine tissue.

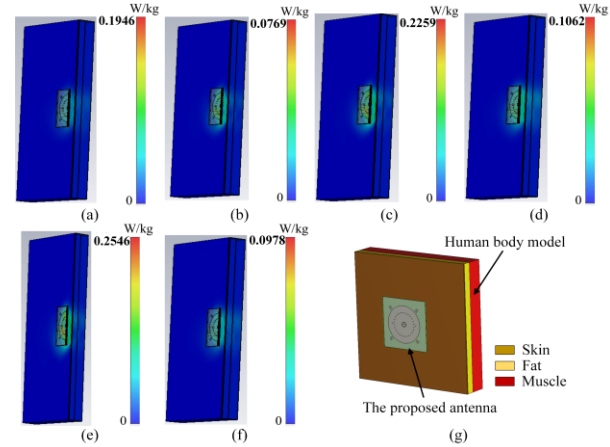


Fig. 10. Simulated SAR. (a) 10g of State 1. (b) 1g of State 1. (c) 10g of State 2. (d) 1g of State 2. (e) 10g of State 3. (f) 1g of State 3. (g) Human model. feeding. It exhibits more controllable radiation patterns and higher gain, which effectively ensures the reliability of on-body and off-body communications.

IV. CONCLUSION

In this work, a polarization and pattern reconfigurable wearable antenna for the 3.5 GHz WiMAX band is designed. The feeding network excites the TM_{01} and TM_{11} modes of the top circular patch separately. Shorting vias are loaded to unify the operating frequencies of the two modes, realizing pattern reconfigurability. Parasitic metal sheets on the top layer realize polarization reconfigurability by controlling their connection state with the circular patch. Measured results of the antenna prototype agree well with simulated ones, and ultra-low SAR values verify its feasibility for wearable applications. The antenna achieves excellent polarization and pattern reconfigurable performance via rational regulation of different mode frequencies and polarization redirection, providing a new design idea for such reconfigurable antennas.

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